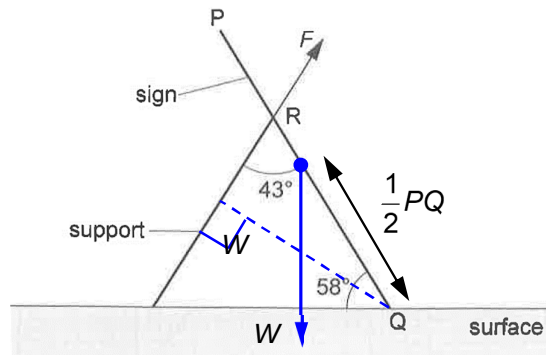


bounces down, hence incorrectly reaching two bounces as an answer.

3(a)



Taking moments about point Q,

$$F \times \frac{2}{3}(PQ \sin 43^\circ) = 2.3 \times 9.81 \times \frac{1}{2}(PQ \cos 58^\circ)$$

$$F = 13.15$$

$$\approx 13.2 \text{ N}$$

Note: The main difficulties to note are in resolving forces F and W to obtain the component that is perpendicular to the line PQ or to determine the perpendicular distance of the line of action of the forces from Q .

3(b)

There is no resultant force acting in the sign in any direction since it is in equilibrium. As the sign is at an angle 58° to the surface, there must be a horizontal leftward frictional force acting on the sign at Q to prevent it from sliding rightward. In addition, the surface also exerts a normal contact force on the sign at Q vertically upwards. Hence, the resultant of the normal contact force and the frictional force on the sign at Q is not be vertical.

No.	Solution
4(a)	By Newton's law of gravitation, gravitational force between two point masses M